

Discourse ergativity and human reference in Basque

Abstract

One possible usage-based motivation for ergative alignment in grammars is the reference-establishing function shared by the sole argument of intransitives and the patient argument of transitives. Here we test this hypothesis, known as Preferred Argument Structure, against discourse data from Basque, a language with consistently ergative alignment in case marking. Drawing on Pear Film retellings our study confirms the discourse-ergative distribution of lexical forms of reference. However, our results also point to animacy as a relevant factor in addition to role as a central driver of this distribution, thus challenging the link between ergative role alignment and reference establishment. This opens new avenues for exploring the specific effects of reference management on grammar. In drawing on production data from a language with consistent ergative marking, we also complement previous studies of discourse ergativity that were based primarily on languages with accusative alignment.

1 Introduction

Humans attend to countless events in their daily lives and about some of them they will talk about with others. Verbalizing events for the purpose of communication includes a phase of event construal and its linguistic realization (Levelt, 1989), and it thus involves the mapping of event participant structure onto syntactic argument structure. This linking between event construal and syntax has been at the center of theoretical and comparative linguistic research for a long time (e.g., Bickel, 2011; Bresnan, Asudeh, Toivonen & Wechsler, 2015; Dowty, 1991; Foley & Van Valin, 1984; Marantz, 1984; Van Valin & LaPolla, 1997).

In this paper, we are concerned specifically with the embedding of event expression in the discourse context during communication. This aspect of event verbalization has been of major interest within usage-based approaches

(cf. Du Bois, 1987; Givón, 1979, 2001) that identify information management as a major driver in sentence production during communication (cf., e.g., Bickel, 2003; Hörberg & Sjons, in press; Stoll & Bickel, 2009) and in the diachronic development of grammatical relations in sentence structures.

The mapping from event participant structure to syntax primarily depends on the number of participants and the argument roles entailed in the meaning of a specific verbal predicate (Dowty, 1991; Van Valin & LaPolla, 1997). Comparative research has focused on cross-linguistic differences and commonalities in the encoding of roles and its alignment across diverse languages, and mostly within isolated (decontextualized) sentence structures. The main function of argument encoding in this view is discriminatory: case marking/flagging, cross-referencing, and (rigid) word order first and foremost serve to differentiate agentive (A) and patientive (P) roles (Comrie, 1981; Dixon, 1979), and various differential marking patterns have been argued to achieve this in the most efficient way (Haspelmath, 2021). Discrimination of roles has also played a role in accounting for interactions between means of argument encoding, for instance between case marking and word order, and it has been suggested in the literature that word order can ‘step in’ where no case marking provides sufficient cues (cf. Levshina, 2019, 545 for discussion and further references).

A role-discriminatory function of argument encoding is, however, general across alignment systems (Comrie, 1981; Dixon, 1979), and thus it can hardly explain cross-linguistic trends in this domain. Since at least the 1970s, the functionalist tradition has focused on the interconnections between syntactic structures and discourse structure. Events expressed successively in discourse are not simply lined up, but are instead interrelated with one another through various cohesiveness ties, e.g., temporal relations (that one event follows another in time) or causal relations (that one event results from another), etc. (Givón, 1983; Halliday & Hasan, 1976; Kehler, 2002). Event participants likewise represent such ties (Halliday & Hasan, 1976), in that these are identifiable with discourse referents (Karttunen, 1976), i.e., representations of entities in the discourse universe that can be referred to successively.

Referential ties arise where interlocutors track the same discourse referent over multiple events expressed across successive sentences. In addition however, discourse referents will need to be introduced into the discourse before they can be tracked. Both referent tracking and introduction involve referential choice, i.e., a choice between different options among the referential forms of a language, e.g. full noun phrase vs. demonstrative or pronoun vs. zero anaphor. While the introduction of new referents plays a pivotal role, both tracking and introduction involve the same parameters of attention

management: speakers monitor the awareness state of their addressee in making such choices so as to ease referent identification (Chafe, 1976,9; Seifart, Strunk, Danielsen, Hartmann, Pakendorf, Wichmann, Witzlack-Makarevich, de Jong & Bickel, 2018; cf. Arnold, 2008 for an overview of this research tradition). This awareness state of referents at a given point in the discourse is commonly called “information status”, and it is crucially affected by factors such as the newness of a referent, as well as various endophoric factors like anaphoric distance, the form and role of its antecedent mention, or the persistence in subsequent discourse.

In principle, role mapping and referential choice are mutually independent processes in discourse production. Any form of reference can appear in any syntactic argument function, reflecting whatever information status a discourse referent bears at a given point in a discourse. In this view, any referential choice is equally likely in any role. However, similar to associations between argument roles and the ontological class memberships of discourse referents filling them (“cast” and “role” associations: Evans, 1997), there are strong correlations between syntactic argument functions and information status (hence, ‘Role and Reference Grammar’, as developed by Foley & Van Valin, 1984; Van Valin & LaPolla, 1997). For instance, subjects in English have been found to host primarily highly accessible referents realized by (unstressed) pronouns or zero anaphors (Prince, 1981), following what Chafe (1994) has termed the Light Subject constraint.

These and similar observations from other languages have given rise to the idea that, in addition to patterns of argument encoding (through case-marking, agreement, or word order), roles enter further types of grammatical relations that capture commonalities in behavioral properties, in particular reference tracking cross sentences (Foley & Van Valin, 1984; Halliday & Hasan, 1976). Under this view, a grammatical relation like ‘subject’ does not only capture the joint marking properties of the S role (the sole argument of intransitives) and the A role (the most agentive argument of transitives), but also their joint function as the host for continuing topics (Givón, 1983) and for anchoring them for coreference and discourse cohesion (Chafe, 1976; Grosz, Joshi & Weinstein, 1995). Pronominal or zero subjects are preferably interpreted as co-referent with a preceding subject, a tendency which is occasionally grammaticalized into rules of conjunction reduction (Bickel, 2011). Similar tendencies have been found in a range of diverse languages (cf., e.g., Givón, 1983, and contributions therein).

Structural choices during discourse production are thus not merely a matter of mapping roles to syntax, but crucially involve questions of information packaging that enhance the integration of any new sentence into the common ground in a given discourse context. This notion is the rationale behind

traditional grammatical relations like ‘subject’, as well as alternations in the associations of syntactic argument functions and roles through lexical (cf., e.g., Filmore’s “buy”, “sell”, “pay”) and grammatical choices (like voice constructions, cf. Foley, 2007, for an overview). The underlying functional basis of grammatical relations is thus to be sought in their discourse functions (cf. Givón, 1979, 2001). Conversely, discourse functions like reference tracking have been regarded as the functional motivation for the diachronic development of grammatical relations (Dixon, 1979; Givón, 1976; Shibatani, 1991). On this view, the joint marking properties of the {S,A} equivalence set emerge as grammatical structures constituting a grammatical relation ‘subject’ under frequent verbalizations of continuing topics in these two roles, in part due to their convergence with animacy and agentivity (Evans & Levinson, 2009a), in the same way that other grammatical structures have been regarded as emergent from discourse patterns (Givón, 1979; Hopper, 1998).

Ergative alignment has posed considerable challenges to these ideas. For example, it has been suggested that in the Australian language Dyirbal – where case marking of full NPs shows ergative alignment, S and P being unmarked (absolutive or nominative case) and A taking overt (ergative) case marking – the unmarked roles S and P represent a syntactic pivot and the anchor for co-reference processing, analogous to subjects in accusative alignment (Dixon, Anderson, Huddleston et al., 1972; cf. Tsunoda, 2011 for analogous findings in Warrongo; Heath, 1979; Schmidt, 1985 for critical assessment of ergativity in Dyirbal). The absolutive {S,P} set of ergative syntax thus appears to resemble a mirror image of subjects, {S, A}, in accusative syntax, both reflecting topic continuity. However, effects of ergativity like in Dyirbal seem exceptional (Anderson, 1976; Dixon, 1979) and most languages with ergative alignment group S and A as the preferred roles in which to express topics and track their coreference.

An alternative approach to ergativity identifies the absolutive grammatical relation with focality rather than topicality, identifying its discourse function with **reference establishment**, i.e., referent introduction and/or (re-)activation (Chafe, 1976,9), rather than reference tracking (topic continuity and coreference interpretation). Under this approach, the absolutive grammatical relation constitutes the grammaticalization of this discourse function of reference establishment (Du Bois, 1987). This notion has come to be known as **discourse ergativity** (Du Bois, 1987)’s 1987 and rests on the hypothesis that discourse universally converges on what is called a ‘preferred argument structure’ (Du Bois, 1987), associating new and lexical information with the S and P roles.

While Dixon et al.’s 1972 reports on syntactic ergativity in Dyirbal have spurred extensive research into discourse production in other languages with

ergative alignments (cf. Section 2), it remains largely unknown to what extent discourse functions like reference establishment and/or reference tracking vary with alignment. In fact, most studies on preferred argument structure have been conducted on languages that do not exhibit – or hardly exhibit any – ergativity in their grammars (cf. Haig & Schnell, 2016). It is thus an open question whether ergative grammar is particularly adapted to the communicative needs of referent establishment in the same way that accusative grammar has been regarded as particularly adapted to the task of referent tracking. The study reported in this paper is intended to fill this gap. We examine the distribution of full (lexical) noun phrases throughout individual texts and across S, A, and P in Basque, a language with ergative alignment in case marking and some control constructions (Berro & Etxepare, 2017; Etxepare, 2003; Laka, 2006; Zúñiga & Fernández, 2021). Crucially, in our textual data from Basque the probability for both S and P to host full lexical NPs (with new or low-accessibility reference) is found to be higher than for A, in line with the preferred argument structure account. Yet, we find little support for a specific reference-establishing function of S and P roles. Instead, distributions of lexical forms follow nearly entirely from the association of human reference with agentivity and high discourse topicality (and hence on-average high referent accessibility). This is in line with findings from Haig & Schnell (2016) who find that human discourse referents show an overall low probability of taking lexical forms of reference, presumably due to their high degree of overall topicality.

2 Ergativity in discourse and grammar

2.1 The discourse functions of grammatical relations

As mentioned above, the treatment of ergative alignment in relation to discourse functions of grammatical relations like ‘subject’ has been notoriously challenging. Earlier research on Dyirbal (Dixon et al., 1972) finds that S and P behave jointly like S and A, representing a mirror image of subjects. It is thus {S, P} that hosts highly accessible referents and continuing topics and functions as the pivot for co-reference interpretation. Hence, zero anaphors in conjunction reduction constructions are restricted to S and P (Dixon, 1972, p. 147-70) in the same way that they are restricted to S and A in a language like English. The following examples contrast English subjects with Dyirbal absolutes (Dixon, 1972):

- (1) a. *The woman_k saw the man_i and $\emptyset_k$ returned.*

b. *nguma_i yabu-nggu_j bura-n Ø_i banaga-nyu*
 father.ABS mother-ERG see-NFUT return-NFUT

‘The woman saw the man and returned.’ [The man returned]

Analogous claims have been made for Warrungu (Tsunoda, 1986). However, since at least Anderson (1976) it has transpired that in many other languages with ergative alignment, co-reference interpretations are chiefly associated with S and A, not S and P. These languages thus have a subject-like ($\{S, A\}$) grammatical relation in terms of their discourse use. Even for Dyirbal, subsequent analyses suggested that discourse topics might not select S and P as much as originally suspected (Heath, 1979; Schmidt, 1985).

An alternative view has been proposed under the heading of ‘preferred argument structure’ (PAS, Du Bois, 1987), where ergative alignment is regarded as a converse to accusative alignment so that $\{S, P\}$ are open to host non-topical, often new, referents and full NPs as their forms of expression. The ‘absolutive’ thus bears the distinctive function of a ‘staging platform’ for referents of low accessibility. The PAS hypothesis is supported by findings from a corpus study of textual data from the Mayan language Sakapultek (Du Bois, 1987), but it has been argued to reflect a universal pattern in discourse production. Under this view, the difference between $\{S, P\}$ and $\{S, A\}$ alignment is a universal difference in function (reference establishment vs. reference tracking), not a difference in how one function (reference tracking) is variably projected into grammatical relations. In other words, the two crosslinguistically most common alignment types reflect two distinct discourse functions and both of these are fundamental to any discourse production (Du Bois, 1987; Evans & Levinson, 2009b). This view has been enormously attractive and has made its way into introductory textbooks (e.g., Croft, 2022, p. 249).

2.2 Accommodating discourse referents

Du Bois’s (1987) account of ergativity in terms of discourse functions draws on the (postulated) necessity for interlocutors to establish discourse referents in the universe of discourse before these can be taken up again over the course of an interaction. As recipients are already occupied with the processing of incrementally incoming information about new events and their interrelations, as well as the tracking of discourse referents figuring as event participants, new referents would disrupt the flow of information to be processed. To avoid any processing difficulties, producers are hypothesized to distribute new referent information in such a way that it will not interfere with other aspects of discourse content. Hence, they will avoid accumulating

	LEXICALITY	NEWNESS
QUANTITY	Avoid two lexical arguments	Avoid two new arguments
ROLE	Avoid lexical A	Avoid new A

Table 1: Preferences in argument realization according to PAS

new referents in single informational units, a pattern captured in Chafe’s (1987) “one new idea at a time”. Related to this is Lambrecht’s (1994, 184–7) Principle of Separation of Role and Reference (PSRR) which states that identification event participants with discourse referents is to be established before their participatory role in the event is verbalized.

While a range of constructions like left-dislocation or presentational constructions can be levied for reference establishment (cf. Prince, 1992, Lambrecht, 1994, 176–84), Du Bois (1987) relates this function to ergative alignment: as producers avoid placing two new referents in a single clause unit, they tend to relegate these to either P or S roles and to actively avoid the A role. By analogy, the placement of full lexical noun phrases (NPs) like “the man” (as opposed to the pronominal form “he”) is restricted to one per clause, and to the roles S or P.¹ Du Bois regards these patterns as reflecting underlying (soft) constraints on discourse production which are summarised in (1).

These preferences were initially observed in the Mayan language Sakapultek, and later confirmed for other languages (Du Bois, 1987, 2003a). The following examples in 2 from English Pear stories (Chafe, 1980) illustrate preferred and dispreferred chains of sentence structures in discourse production:

- (2) ENGLISH (cf. (Chafe, 1980, 301, 316))
- a. [.25] *there was*- [.25] *a man*, [.2] *who was picking pears* preferred
 - b. [4.65]] *a man was picking pears* dispreferred

While both sentences are attested (and grammatical), the version in 2a is more common in language usage as it abides by the PAS constraints. This example suggests that a central role for discourse processing is borne by intransitive constructions, particularly those with what Du Bois (1987) characterizes as ‘semantically bleached’ predicates like ‘come’, ‘appear’ or ‘arrive’,

¹In later work, Du Bois (2003b, 49) focuses entirely on lexical NP use as a proxy for low accessibility of referents. Therefore, the placement of lexical NPs in either the S or the P (i.e., the absolutive) is motivated by the low accessibility of their referents, which interlocutors are stipulated to process more readily in this way.

as well as presentational constructions such as English *there is*. As these do not contribute any further event-related information to the discourse, they primarily serve purposes of reference establishment, and hence the processing of referents does not interfere with the remaining content of a discourse. Producers would thus deploy intransitive sentences with presentational function deliberately for the sake of referent introduction (or ‘reactivation’) where necessary, in particular to avoid the accumulation of low accessibility referents.

The role-related constraint on A as a host for low-accessibility referents with lexical form, however, does not follow from the quantity-related constraint. Du Bois (2003b) makes it clear that the discourse-ergative distribution of lexical/new NPs is not merely a reflection of the constraint against the culmination of two such expressions in a single transitive clause. Rather, according to Preferred Argument Structure, it is the specific function of absolutive NPs to accommodate lexical/new arguments so that these NPs come to host information that is more cumbersome to integrate into their discourse model than other information. Hence, Du Bois characterizes the absolutive as a ‘predictable locus for unpredictable work’, the latter referring to the unpredictability of referents that are not continuing topics. The A role is avoided by speakers for lexical/new NPs because addressees will not expect such information there. Hence, while subjects serve to regulate coreference interpretation of continuing topics, the absolutive serves the converse purpose of establishing new or less accessible referents to be taken up as topical in subsequent discourse.

As the discourse-ergative distribution of new/lexical NPs follows, according to Preferred Argument Structure, from a need to mitigate against processing costs on behalf of the addressee/recipient, Du Bois (1987) postulates that it will materialize only under conditions of high information pressure, i.e., a high density of discourse referents in a given (stretch of) discourse. For instance, Du Bois’s 1987 original corpus from Sakapultek exhibited very high information pressure, with a new referent being introduced nearly every second clause on average, whereas the Pear stories from English (Chafe, 1980 investigated by Kumagai, 2006) had a new referent being introduced only approximately every seven sentences. According to Du Bois (1987), it is the S role that is sensitive to information pressure. Under conditions of low information pressure, the levels of lexical/new NPs will be relatively low in S, and under conditions of high information pressure, they rise to levels comparable to those in P, hence yielding the discourse-ergative signature. Under the Preferred Argument Structure hypothesis, the absence of discourse ergativity in the English Pear stories found by Kumagai (2006) would thus be explained by the low information pressure of this corpus. As we will show below, the S

role is indeed of particular interest in the context of our study, yet in ways quite deviant from those postulated in Preferred Argument Structure.

2.3 Challenges for the theory of discourse ergativity

Its theoretical appeal notwithstanding, more recent evidence has challenged parts of the discourse ergativity hypothesis. Most importantly, several studies show that S arguments do not universally align with P in terms of their lexicality (Everett, 2009; Haig & Schnell, 2016; Haspelmath, 2006; Schnell, Schiborr & Haig, 2021). Instead, S typically occupies an intermediate position between A and P, yielding a tripartite rather than an ergative pattern. For the two corpora in Haig & Schnell’s sample that did exhibit discourse ergativity, other factors such as extremely high information pressure (in Sakapultek Pear Stories) and repetition of lexical NPs in S (in English classroom interactions) were identified. Discourse ergativity is thus not a general property of discourse but may rather occur under special circumstances.

By contrast, low lexicality of A arguments is found repeatedly in corpora from diverse languages. However, rather than reflecting specific structural properties of this syntactic position and any constraints associated with it, the low lexicality of A arguments resembles an epiphenomenon of animacy and topicality considerations (Haig & Schnell, 2016). Agentive roles are more likely to be filled by human participants, which are in turn more discourse-topical and hence more likely to receive a reduced form of expression rather than a full lexical NPs. The S role is intermediate in levels of lexicality because it is less restricted in terms of role mapping. It thus admits nonhuman or inanimate referents, which are often also nontopical and hence less accessible, attracting lexical form of expression.

Furthermore, it seems impossible to establish from corpus data alone that the occurrence of lexical NPs in the S role is ultimately motivated by considerations of information management on behalf of the addressee: Schnell et al. (2021) suggest that intransitive sentences with new referents in S are neither semantically bleached nor obviously motivated by considerations of information pressure. Overall, such intransitive sentences with new referents resemble narrative sentences that contribute to the development of a discourse but which happen to bear a new referent.

Yet another concern is that most research into preferred argument structure has drawn on discourse data from languages with accusative alignment and so it is premature to assess it on a global scale. It may well be the case that preferred argument structure varies by the dominant alignment in a language. In particular, effects for discourse ergativity might be stronger

in languages with consistent ergative case marking.

Finally, we note that the use of lexical arguments varies substantially across cultures (Bickel, 2003; Duranti, 1994). For example, studies on lexical referential density have looked at this by tracking the ratio of overt NPs (lexical and pronominal) over the total number of argument slots (Stoll & Bickel, 2009). More specifically, Stoll & Bickel investigated the temporal dynamics of lexical reference in culturally and typologically varied languages, looking at the narration of Pear Stories retellings by speakers of a language without ergative case (Russian) and speakers of a language with ergative case (Belhare). They found that, while Russian narrators seek to introduce explicit forms of reference from their initial mention, Belhare speakers tend to leave the use of explicit forms to a later point in the narrative where discrimination of participants becomes more urgent. While it remains unclear whether the differences are related to alignment type or cultural traditions in story-telling, it seems critically important to assess lexical reference in its temporal dynamics rather as a grand average of discourse.

For these reasons, we will look at the lexical referential density of both intransitive and transitive structures throughout the discourse, paying special attention to the lexicality constraints by PAS and comparing our results to the ones by Stoll & Bickel (2009), to see whether the alignment type can have an effect on where throughout the discourse lexical arguments are used. Given the ergative nature of Basque, we will compare our results to those of Stoll & Bickel to see whether Basque follows the lexical referential density of Belhare, or whether the lexical referential density is independent of alignment type and would instead reflect cultural patterns. We suspect that cultural variation would lead Basque speakers to behave more like Russian than Belhare speakers.

3 Basque as a test case

Research into PAS and discourse ergativity has to date nearly exclusively drawn on languages with mostly accusative alignment (e.g., Haig & Schnell, 2016). PAS studies on languages with non-accusative alignment are comparatively rare, see e.g. Brickell & Schnell (2017) on the symmetrical voice language Tondano (Sulawesi) and Matter (2020) on Mapundungun, a language with a direct-inverse system. In response, we turn to a language with consistently ergative case-marking, testing whether such marking enhances discourse ergativity and whether this modulates how speakers establish reference.

Basque presents itself as an ideal test case for this purpose. Ergative case

marking is used for agents in both transitive and monovalent verbs, and it encompasses the entire nominal lexicon, thus including personal pronouns of all persons, as well as nearly all tense, aspect, and mood (TAM) categories, except for the progressive aspect (Laka, 2006). It also possesses an antipassive construction (Zúñiga & Fernández, 2021). Hence, in any transitive sentence in active voice with overt core arguments, the agentive role is assigned ergative case, and the patientive role absolutive case. The S role in an intransitive sentence is assigned the ergative if it is agentive and absolutive if it is patientive:

- (3) *katu-ak txori-a ikus-ten d-u-∅*
cat-ERG.SG bird-ABS.SG see-IPFV 3SG.P-AUX.PRS-3SG.A
‘the cat sees the bird’
- (4) *katu-a lokar-tu d-a*
cat-ABS.SG fall.asleep-PFV 3SG.S-AUX.PRS
‘the cat has fallen asleep’
- (5) *katu-a-k salto egin d-u-∅*
cat-SG-ERG jump do 3SG.P-AUX.PRS-3SG.A
‘the cat has jumped’

Our main purpose in this study is to explore the possibility that ergativity in Basque grammar may yield ergative constellations in the realization of arguments during discourse production. As most previous studies – and in particular those yielding substantial counter-evidence to the PAS hypothesis – have been based on languages with exclusively accusative alignment, e.g. Everett (2009) and Haig & Schnell (2016), it is possible that languages that have developed ergative alignment in their argument encoding systems are also more prone to show discourse ergativity.

4 Materials and Methods

In this section, we motivate our decision to use a Pear story corpus and briefly describe its features and how it was annotated (4.1). We then turn to explaining the statistical procedures deployed in our analyses (4.2).

4.1 The Basque Pear Film retellings

For our study of PAS in Basque, we draw on a corpus of Pear Film retellings (Aske, 1998). The Pear Film is a silent movie that is close to six minutes long and was designed as a stimulus for comparative linguistic research (Chafe,

1980). The movie revolves around an older man who is picking pears in a pear grove and a boy who steals one of his baskets full of pears. The boy rides the pear basket on his bike along a dust road, and upon meeting a girl coming his way, he runs into a rock and falls, the pears spilling on the ground. A group of three children helps collect the pears, and the pear thief gives each of them a pear as a reward. He rides on and the group of boys walks in the direction where the boy came from, thus passing the older man while eating their pears. This marks the end of the story.

The Pear stories are elicited in the following experimental setup: the movie is shown to one person who is asked to watch and memorise the content of the story shown. They can choose to watch the movie a second time. In a second step, the experimenter brings in a further person who speaks the same language or the variety of interest. The first person is then asked to recount the story depicted in the movie as the second one – so it is explained – is not familiar with it. In the case of the Basque Pear Stories, the listener was a school or work colleague of the narrator (Aske, 1998, 262).

The merit of the Pear Film as a stimulus is that it is void of any linguistic stimulus and is hence suitable for comparative linguistic research. Moreover, the use of a stimulus for discourse elicitation ensures at least some level of content control. A further advantage of Pear Stories (as the retellings are commonly known as) is that they trigger discourse production that is heavy on third person reference and nearly void of reference to speech-act participants. This avoids any complication through person deixis and hence enables focused investigations into third-person anaphoric versus discourse-new mentions and their forms of expression, i.e., lexical vs. nonlexical.

A further relevant aspect is that Pear story corpora have been used in both PAS research (Du Bois, 1987; Kumagai, 2006) and in research on referential density (Bickel, 2003; Stoll & Bickel, 2009).

Our analysis is based on the transcriptions of the original recordings of the spoken retellings, which jointly amount to 1702 utterances (with each speaker producing between 39 and 125 sentences). The corpus was manually annotated following an approach similar to Bickel (2003). Each clause that included a verb was annotated for the total number of arguments and the number of realized arguments. From the total number of realized arguments, whether the arguments were pronominal or lexical was annotated (thus distinguishing between zero anaphora on the one hand and between pronominal and lexical on the other). Then, at each verb, whether there was an A, S, or P was annotated, also annotating the corresponding lexical word each role in the sentence was assigned to (regardless of whether the word was present, pronominalized or a zero anaphora).

For the analysis, utterances that did not include a verb and ditransitive

sentences were excluded. Sentences that had a subordinate clause as a subject of intransitive, subject of transitive, or object of transitive clauses were excluded. Moreover, sentences that included the progressive “ari” marker in Basque, which does not involve ergative marking since it removes the agentivity of the single argument in the clause (Laka, 2006; Laka, Coon, Massam & Travis, 2017), were excluded. This was done since this type of structure with a double verb construction are not considered in PAS research. In total, the corpus data underlying the current study consist of 345 transitive and 516 intransitive sentences.

Finally, the corpus contained a small number of non-human animate nouns (seven instances of the Basque word for “goat”). This low number of data points does not allow making statistical inferences. Therefore, non-human animate referents were excluded and only human and inanimate referents were used for the analysis.

4.2 Statistical analysis

All the sentences were grouped by speaker, A vs. S vs. P role, and lexical vs. nonlexical form of reference. We tracked every lexeme from the first time they were realized until they were overtly realized again, tracking the roles where a given lexeme was used and the number of times it was realized as a pronoun or omitted in between each overt form. This grouping, therefore, yielded unique entries per lexeme per role per speaker, adding the total times a role was instantiated by a lexeme out of the total times the role was realized.

Two different Bayesian binomial models were fitted in R (R Core Team, 2022) using the brms interface to Stan (Bürkner, 2017,1; Carpenter, Gelman, Hoffman, Lee, Goodrich, Betancourt, Brubaker, Guo, Li & Riddell, 2017). The first model pertained to the probability for each role (A, S, or P) to receive lexical form of reference, independent of animacy. The second model takes into account the interaction of the role and its referent’s animacy ([+human] and [+inanimate], i.e., A/+human, S+human, S/+inanimate, P+human, P/+inanimate), to receive lexical form of reference. Since there was a small number of inanimate agents that did not allow making statistical inferences on the A/+inanimate condition, these were removed in order to fit the second model. Treatment coding was used for the predictors in both models. The speaker and the lemma were added as random intercepts to both models, respectively.

Both models were fitted in four chains of 9000 iterations each (including 4000 warmup iterations). We used normal priors with $\mu = 0$, and $\sigma = 10$ for the predictors, and half Student- t priors with $\mu = 0$, $df = 3$, and $\sigma = 2.5$ for the random effects.

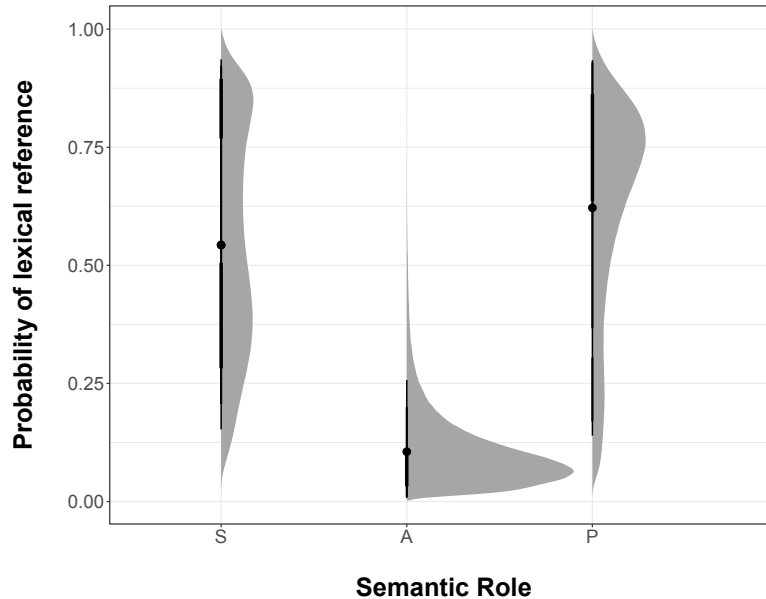


Figure 1: Posterior predictions for the use of lexical NPs in the corpus for each role. The dot represents the median and the bars represent the 50%, 90%, and 95% highest density intervals (Kruschke, 2015), respectively.

5 Results

In reporting the results of our probabilistic modeling, we will first present the probabilities of individual roles A, S, P to be lexical; our findings here lend initial support to the original PAS hypothesis. We then turn to the model that includes the human vs. inanimate distinctions. This model, combining role and animacy, provides more insight into the choice of lexical NPs and thus challenges PAS theory. Finally, we report probabilities for S in intransitive sentences vs A or P in transitive sentences to be lexical over the time course of each of the 22 narratives; models suggest the greatest probability for S to be lexical in narrative-initial sentences. This suggests that – contrary to PAS – S is not specifically deployed to mitigate against hypothetical processing challenges under high information pressure, since this pressure increases only later in the Pear Stories (Stoll & Bickel, 2009). In 6 we discuss our results in relation to the original PAS hypothesis and more recent critical work dealing with PAS.

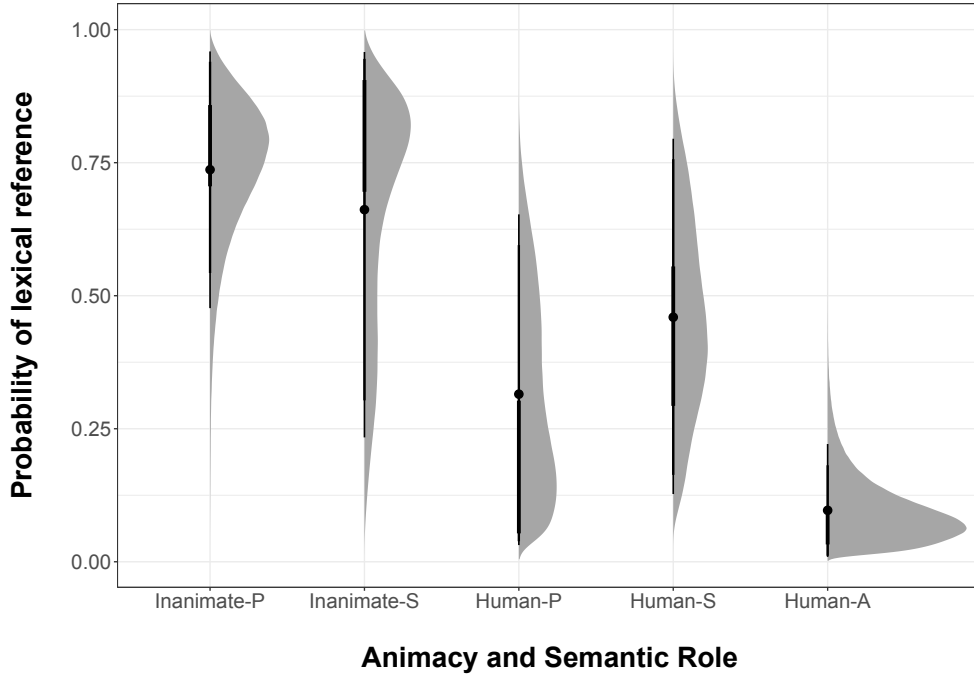


Figure 2: Posterior predictions for the use of lexical NPs in across the corpus by animacy and role. The dot represents the mean and the bars represent the 50%, 90%, and 95% highest density intervals (Kruschke, 2015), respectively.

5.1 Modeling the impact of A, S, P only

Figure 1 shows the posterior predictions for the model with role (A, S, P) as the only predictor. The posterior distribution plot shows that the median values for S and P are very close to one another, with both values being around 50%, whereas the probability for lexical As is much smaller. The odds for lexical A is less than 1/7 of those for lexical S (and an odds ratio < 1 has a 99% posterior probability), while the estimated difference between lexical P and lexical S includes 0 (see Table A1).

5.2 Modeling the impact of A, S, P interacting with animacy

Figure 2 shows the results of the model with roles (A, S, P) combined with the animacy distinctions (human, inanimate; see Table A2). The choice of lexical form of reference in Basque is strongly correlated with both animacy and role. Inanimate NPs are overall more probable to be realized lexically. The odds for inanimate S to be lexical are 3.86 times the odds for human S

to be lexical (with 99% probability that the odds ratio is larger than 1); the odds for inanimate P to be lexical are 21.54 times the odds for human P to be lexical (with more than 99% probability that the odds ratio is larger than 1). For human NPs, the odds for S and P are 7.46 and 3.29 times the odds for A roles to be lexical, respectively (both with more than 99% probability that the odds ratio is larger than 1). In addition, the odds for inanimate NPs in the P role are 2.48 times the odds for NPs in the S role to be lexical (with 92% probability that the odds ratio is larger than 1). For human NPs, however, the odds for the S role to be lexical are 2.25 times the odds for the P role to be lexical (with 99% probability that the odds ratio is larger than 1).

5.3 Distribution of lexical NPs throughout each narrative

In addition to role and animacy as factors impacting the choice of lexical forms of reference, we investigated the Basque Pear stories for the distribution of lexical noun phrases throughout each narrative. PAS suggests lexical forms in S in particular when the mentioning of different discourse referents culminates, and hence information pressure is comparatively high. From Stoll & Bickel’s (2009) study, information pressure emerges as highest about halfway into each narrative and relatively low at its beginning. Hence, given the virtual lack of any information pressure at the beginning of a narrative, the mitigating function of S should be less relevant, and S should not differ in its propensity to take lexical form from the A and P roles.

The distribution of intransitive sentences with a lexical S across the Pear Stories corpus is given in Figure 3. The S role is most likely to take lexical form at the very beginning of each narrative; the use of lexical NPs in S has a peak of about 60% at the very beginning, and then drops by half about 1/4 into the story. The use of lexical NPs then becomes much more stable, shifting between 30% and 40% through to the end of the story, with a second peak after about 3/4 into the story.

When applying again the animacy distinction between human and inanimate referents within the S role (Figure 4), we find that the rates of lexical expression for human referents are highest at the beginning of each narrative and drop from an initial 63% to just above 25% within the first 1/5 of each narrative, followed by a fairly steady ratio of 25% to 38% for the rest of each narrative. Inanimates, on the contrary, show a fairly steady rate of just above 62% for 3/4 of the story, and no inanimate referents appear in the last quarter of the narrative.

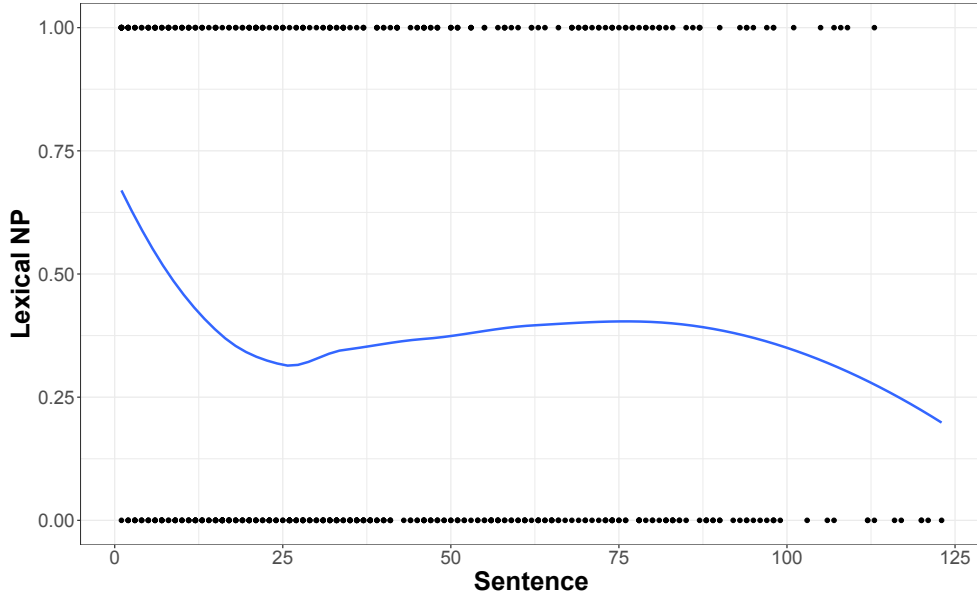


Figure 3: Generalized additive model fit showing smoothed probabilities of lexical NPs used in intransitive sentences across the whole corpus.

This being said, we should note that there are many more datapoints for humans overall than for inanimates regardless of their form of reference. The influence of the larger amount of datapoints for humans makes the intercept in Figure 3 take a similar shape to that of Figure 4 (A). However, the higher rate of lexicality for inanimates, shown in Figure 4 (B), makes the intercept in Figure 3 have a higher rate overall when compared to Figure 4 (A). Thus, the panels in Figure 4 show that inanimates have a stronger tendency to be lexical than humans. The intercept in Figure 4 (B) shows that the lexical rate of inanimates is higher throughout each narrative.

The distribution of transitive sentences with lexical NPs throughout each Pear story is shown in Figure 5. The plot shows an intercept for the lexical value of the joint A and P roles (blue line), the value of the A role (gold line), and the value of the P role (red line) as a function of time. Here we can observe that the use of lexical NPs barely occurs for both roles at the same time (mean counts stay below 1), in line with the quantity-related constraint in the PAS theory. In other words, lexical NPs in transitive sentences have a tendency to appear in one of the roles at a time. This is far more often the P role than the A role (see also Figure 2). The highest use rate for lexical NPs in transitive sentences is found at the very beginning of each narrative, in much the same way found in intransitive clauses (Figure 3).

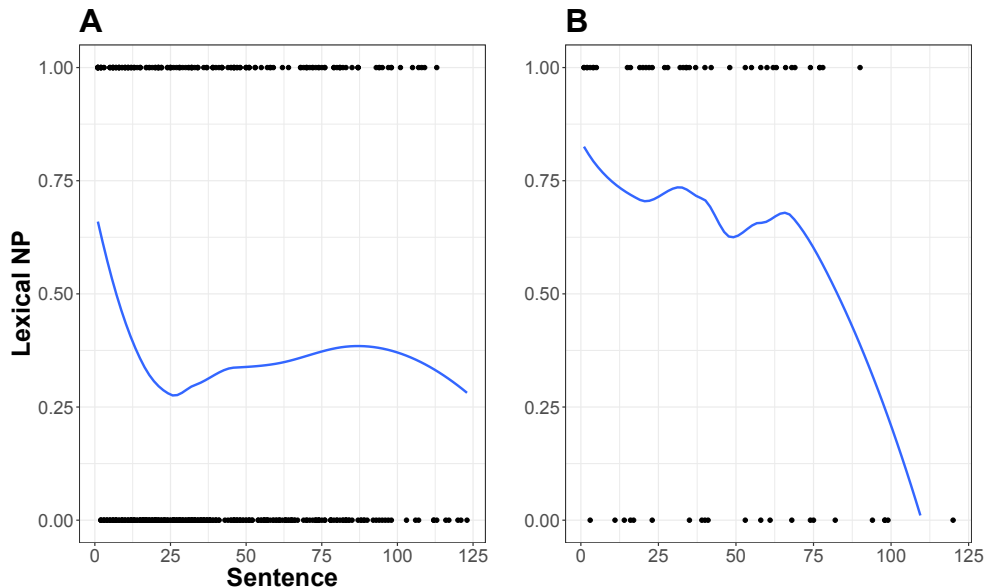


Figure 4: Generalized additive model fit showing smoothed probabilities of lexical arguments used in intransitive sentences across the whole corpus. A: Proportion of human lexical NPs. B: Proportion of inanimate lexical NPs.

6 Discussion

In some respects, our findings match earlier work and support the PAS hypothesis (Du Bois, 2003a,0,1; Everett, 2009; Haig & Schnell, 2016; Haspelmath, 2008): both S and P contrast sharply with A in their higher probability to receive lexical form of reference. Moreover, the overall low probability for A to receive lexical expression squares with the quantity constraint of PAS, i.e., that no more than one core argument should take lexical form and/or host low-accessibility referent.

However, some of our findings are at odds with PAS: firstly, the sharp contrast of S and P with A in lexicality notwithstanding, S shows a smaller probability to receive lexical form than P, thus echoing the tripartite pattern, with intermediated levels of lexicality in S, reported by Haig & Schnell (2016) for the great majority of language corpora they investigate. Hence, our results are consistent with their observation that a shared contrast with A does not entail discourse-functional unity of S and P. Secondly, we find animacy distinctions to be crucial in accounting for the observed distributions of lexical forms. Our analysis shows that the combination of A, S, P roles with animacy distinctions (+human vs. +inanimate) reveals better insights into the use of lexical forms of reference than a model without

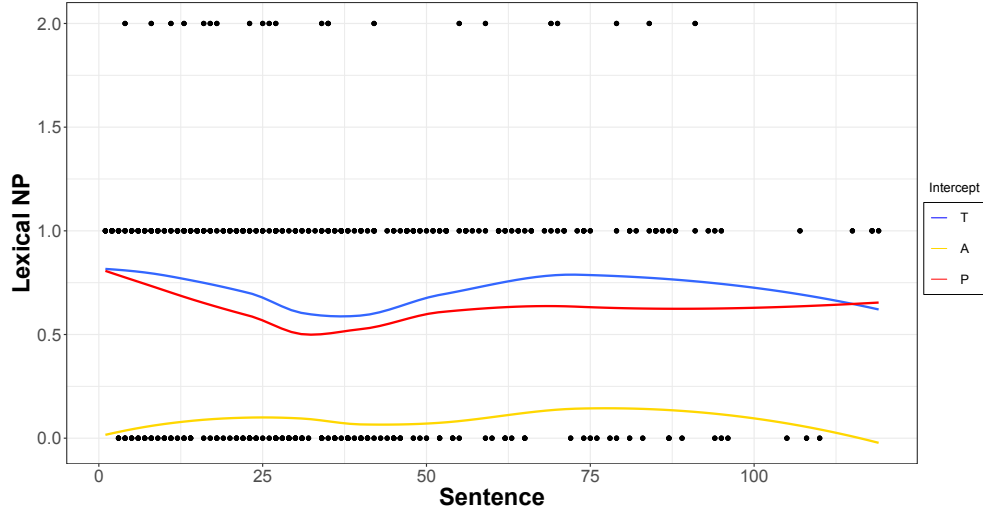


Figure 5: Generalized additive model fit showing smoothed probabilities of lexical NPs used in transitive sentences across the whole corpus (T = transitive sentences; A = all agents lexical; P = all patients lexical).

animacy. In line with Haig & Schnell’s 2016 findings, we find animacy to have the strongest effect on lexical expression: as humans exhibit higher thematic prominence than inanimate discourse referents, they are also more likely to be discourse-topical in the sense of Givón (1983), and most often the topic in a given sentence (Lambrecht, 1994), and thus overall more likely to receive non-lexical expression. In Haig & Schnell’s corpus data animacy in conjunction with discourse-topicality considerations squares with the different lexicality rates in different roles: as agents tend to be human and topical, A is often non-lexical; the reverse holds for P. The intermediate levels of lexicality in S can be taken to reflect its underspecification for role, which may be agentive or patientive. Consequently S overall admits both human and nonhuman or inanimate discourse referents with variable degrees of discourse topicality, and hence overall intermediate levels of lexical forms. While Haig & Schnell (2016) find an animacy split in S with regards to lexicality rates – human S show low rates of lexicality like A, nonhuman ones high rates like P – the Basque data exhibit somewhat higher rates of lexicality for human S arguments. In fact, while the probability for inanimate S to be lexical is smaller than the probability for inanimate P to be lexical, the reverse holds for human S vs. human P where the probability for human S to bear lexical form is in fact greater (with a decisive posterior probability). While we do not have an explanation for this, it does seem to tilt the overall distribution of lexical forms towards discourse ergativity. It is likewise worth noting that the

lexicality rates for human Ps – not investigated separately by Haig & Schnell (2016) – are not as low as those for (human) A either. It seems then that the discourse-ergative distribution attested in Basque narrative discourse is to a major extent driven by animacy, much in line with Haig & Schnell (2016), but also by minimal preferences in the use of lexical forms for human S, and thus potentially for a presentational function of some intransitive predicates.

While further research is needed to probe this possibility, our study challenges information pressure as a major factor. Lexical mentions are by far most likely at a very initial stage in each narrative, where information pressure is naturally maximally low. Lexicality rates do rise somewhat towards the middle phase in each narrative where characters in the story interact (Stoll & Bickel, 2009), but the bulk of lexical mentions, including for the most part lexical S, comes from narrative-initial mentions. It thus appears that the high rates of lexical forms for human S are not motivated by considerations of information pressure. What may still be possible is that some (very short) specimens of Basque Pear stories exhibit what Haig & Schnell (2016) term a ‘telegraphic style’ of discourse production which they suggest may explain the distribution of lexicality in the Sakapultek corpus: in some cases speakers may not be dealing well with the task of re-narrating the movie, and they may thus resort to recounting the appearance of people and things, but not elaborate much on these.

The high lexicality in S, including human S, is potentially grounded in a preference for referent introduction or topicalization (by way of promotion in topic acceptability, Lambrecht, 1994), in intransitive constructions. Differences in that may result from a culturally determined habituation in discourse production. That languages – and cultures – may differ with regards to reference establishment has been reported by Stoll & Bickel (2009), who find significant differences in lexical referential density between speakers of Belhare and Russian that they explain in terms of differences in habits of discourse production, so that Belhare interlocutors have little concern for the exact identity of discourse referents up to the points where identification of different referents becomes crucial so as to avoid compounded difficulties in discourse processing through referential ambiguities. Russian speakers, to the contrary, seek to first establish new discourse referents (and potentially those mentioned a long time ago) before elaborating on these and their involvement with other discourse referents, hence producing overall higher rates of lexical reference compared to Belhare speakers. A further dimension of the concern for reference establishment may thus be reflected in the use of intransitive constructions to host low-accessibility referents, and Basque discourse ergativity may thus be the specific reflection of a particular concern for reference establishment.

The fact that Basque is more similar in these regards to Russian than to Belhare suggests that the effects we found are not driven by ergativity; Belhare is as consistently ergative in its case system as Basque.

7 Conclusion

In this paper we have investigated the distribution of full lexical noun phrases in a corpus of Basque Pear Film retellings with the aim of testing the hypotheses of Preferred Argument Structure as a possible account of discourse ergativity. While we find the discourse-ergative patterns predicted by PAS overall instantiated in our Basque discourse data, we also find minimal differences between S and P and crucially we find no indication that these patterns result from the information-management factors that had been proposed to drive the ergative distribution of lexical NPs. Overall, our results point to animacy as the major driver of nonlexical reference, and, more generally, to an association of different ontological classes with different roles as the main determinant of lexicality distributions across A, S, P, with the provision that human S show unexpectedly high rates of lexical forms.

Data availability

Data and analysis scripts are available from https://osf.io/4svbu/?view_only=e2d6cd03950640248bdb2276d0a2f3ac.

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Appendix

Distribution of lexical NPs in transitive sentences

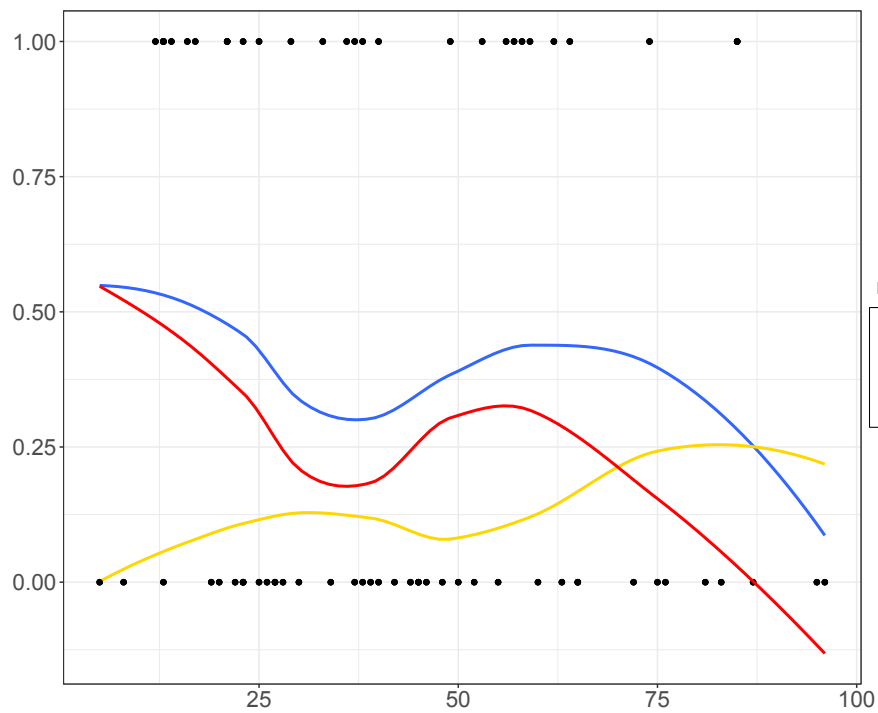


Figure A1: Generalized additive model fit showing the number of lexical NPs used in transitive sentences across the whole corpus where all patients are human (T = transitive sentences; A = all agents lexical; P = all patients lexical).

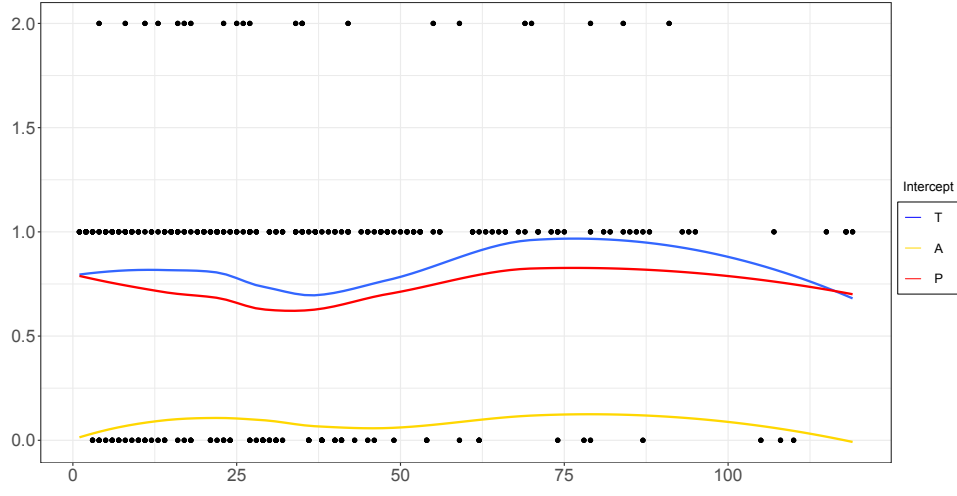


Figure A2: Generalized additive model fit showing the number of lexical NPs used in transitive sentences across the whole corpus where all patients are inanimate (T = transitive sentences; A = all agents lexical; P = all patients lexical).

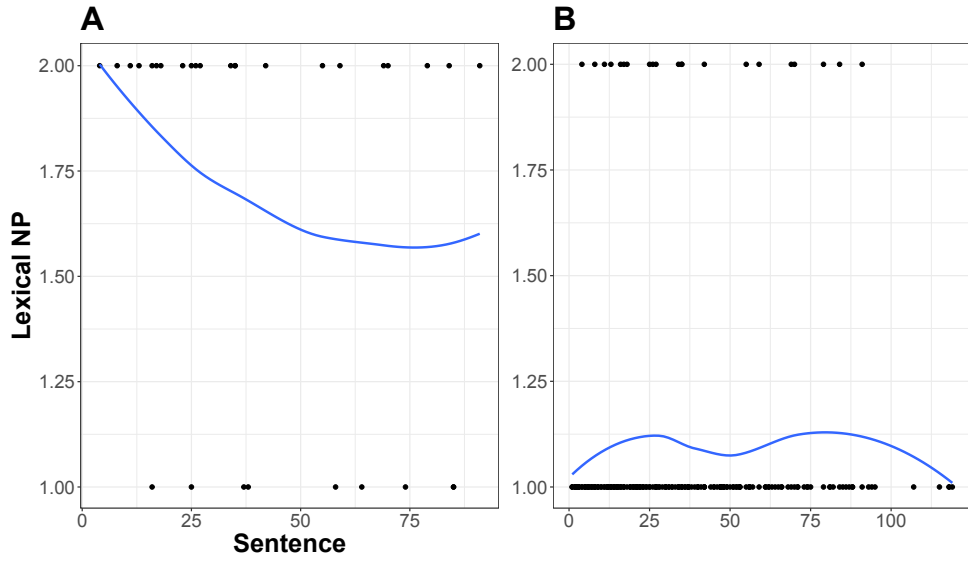


Figure A3: Generalized additive model fit showing the number of lexical NPs used in transitive sentences across the whole corpus. A: $P(\text{lexical}|\text{agent}) = 1$. B: $P(\text{lexical}|\text{patient}) = 1$.

Model summaries

Table A1: Posterior parameter estimates from Bayesian binomial regression predicting lexical NPs for agents (A) and patients (P) compared to sole arguments (S). Coefficients are in log odds differences from the intercept, which represents S (so the odds ratios in the main text are $e^{\hat{\beta}}$). The posterior probability of a parameter is reported when it is at least 90%. We report $P(\hat{\beta} > 0)$ when mean $\hat{\beta} > 0$ and $P(\hat{\beta} < 0)$ when mean $\hat{\beta} < 0$.

Parameter	Mean $\hat{\beta}$	SE	Highest Density		$P(\hat{\beta} < > 0)$
			Interval	Quantiles	
			5%	95%	
Intercept (S)	0.330	0.754	-0.43	1.10	
A	-1.990	0.454	-2.44	-1.54	$P(< 0) > 0.999$
P	-0.426	0.518	-0.93	0.09	$P(< 0) = 0.952$

Table A2: Posterior parameter estimates from Bayesian binomial regression predicting lexical NPs for inanimate S, human P, human S, and human A compared to inanimate P. Coefficients are in log odds differences from the intercept, which represents inanimate Ps (so the odds ratios in the main text are $e^{\hat{\beta}}$). The posterior probability of a parameter is reported when it is at least 90%. We report $P(\hat{\beta} > 0)$ when mean $\hat{\beta} > 0$ and $P(\hat{\beta} < 0)$ when mean $\hat{\beta} < 0$.

Parameter	Mean $\hat{\beta}$	SE	Highest Density Interval Quantiles		$P(\hat{\beta} < > 0)$
			5%	95%	
Intercept (inanimate P)	0.948	0.807	0.170	1.790	
Inanimate S	0.034	0.789	-0.74	-0.84	$P(> 0) = 0.534$
Human P	-2.110	1.160	-3.31	-0.96	$P(< 0) = 0.999$
Human S	-1.300	1.000	-2.34	-0.30	$P(< 0) = 0.991$
Human A	-3.300	1.060	-4.36	-2.21	$P(< 0) > 0.999$